Experiment No. 7
FUNCTIONS, ARRAY, and STRINGS in C++

I. OBJECTIVES

1. To employ different function statements in C++ programs.
2. To differentiate and use standard functions and programmer-defined functions in different applications.
3. To employ different string function statements in C++ programs.
4. To be able to implement the two types of array.

II. LABORATORY EXERCISES

Encode the following program and compile them. Illustrate the output.
Note: if getch(); will not work on other C++ ide compiler replace it with _getch();
For www.onlinegdb.com editor and cxxdroid (mobile app) you may remove it.

Program no.1 (save as functionex1.cpp)

```
#include<iostream>
#include<conio.h>
#include<windows.h>
using namespace std;
void gotoxy ( short x, short y )
{
  COORD coord = {x, y};
  SetConsoleCursorPosition ( GetStdHandle ( STD_OUTPUT_HANDLE ), coord );
}

int main ()
{
// Set text color as Yellow with white background.
SetConsoleTextAttribute(
    GetStdHandle( STD_OUTPUT_HANDLE ),
    FOREGROUND_INTENSITY | 6|// Text color as Yellow.
    BACKGROUND_INTENSITY | BACKGROUND_RED|BACKGROUND_GREEN|BACKGROUND_BLUE );
// White Bg.
      gotoxy(31,12);cout<<"OFFICIAL RECEIPT";
_getch();
return 0;
}
```

Program No. 2 (save as functionex2.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int funct(int x);
int main ()
{
      int c;
      for (c=1; c<=5;c++)
      {
```

```
            cout<<"\n"<<funct(c);
      }
      _getch();
      return 0;
}

int funct(int x)
{
      int y;
      y=x*x;
      return y ;
}
```

Program No 3. (save as functionex3.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int subprog(int c);
int y;
int main ()
{
      int c;
      c=1; y=0;
      while(c<=5)
      {
            cout<<" "<<subprog(c);
            c++;
      }
      _getch();
      return 0;
}
int subprog(int x)
{
      y+=x;
      return y ;
}
```

Program No 4. (save as string1.cpp)

```
#include<iostream>
#include<string>
usingnamespacestd;
intmain()
{
      stringname("My true north");
      stringcrush("Tara my love");
      cout<< name <<"\n";
      cout<< crush;
      cout<<"\nThe length og string name is "<<name.length();
      cout<<"\nThe length og string crush is "<<crush.length();

      name +=crush;   // string
      //name += '\n';           // character
      cout<<"\n";
      cout<<"The new content of string name: "<<name;
```

```
	cout<<"\nThe new length of string name is "<<name.length();

	cout<<"\n"<<crush.append(" of my life.")<<"\n";
	system("pause");
	return 0;
}
```

Program no.5 (save as string2.cpp)

```
#include<iostream>
#include<string>
usingnamespacestd;
intmain()
{
	string word, temp;
	intlen, i, x, upper = 0, lower = 0, number = 0, special = 0;;
	cout<<"Enter any word/paragraph: ";
	getline(cin,word);
	len = word.length();
	cout<<"The reverse form is: ";
	for (x = len - 1; x >= 0;x--)
	{
		cout<<word[x];
	}


for (i = 0; i<len; i++)
	{
		if (word[i]>= 'A'&& word[i]<= 'Z')
			upper++;
		elseif (word[i]>= 'a'&& word[i]<= 'z')
			lower++;
		elseif (word[i]>= '0'&& word[i]<= '9')
			number++;
		else
			special++;
	}
	cout<<"\nUpper case letters: "<< upper;
	cout<<"\nLower case letters : "<< lower;
	cout<<"\nNumber : "<< number;
	cout<<"\nSpecialcharacters : "<< special;

	for (i = 0;word[i] != '\0';i++)
	{
		word[i] = toupper(word[i]);
	}
	cout<<"\nAll caps: "<<word<<endl;

system("pause");
return 0;
}
```

Program No.6 (save as array1.cpp)

```
#include<iostream>
#include<conio.h>
usingnamespace std;
int main ()
```

```
{
int a, b=0;
int c[10]={1,2,3,4,5,6,7,8,9,0};
for (a=0;a<10;a++)
      if ((c[a]%2) == 0)
            b+=c[a];
cout<<b;
getch();
return 0;
}
```

Program No 7. (save as array2.cpp)

```
#include<iostream>
#include<conio.h>
usingnamespace std;
int main ()
{
int r,c;
int num[2][3];
//loads the values
for(r=0;r<2;r++)
      for(c=0;c<3;c++)
            num[r][c]= (r*4)+(c+19);
//displays the previously loaded values
for(r=0;r<2;r++){
      for(c=0;c<3;c++)
            cout<<" "<<num[r][c];
            cout<<"\n";
}
getch();
return 0;
}
```

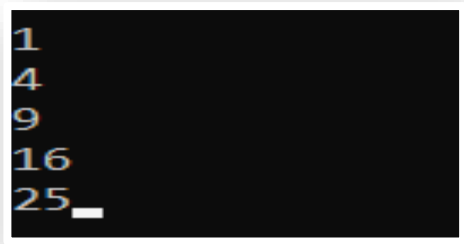
III. Summary of Program Outputs

Direction: Demonstrate the corresponding output for each of the given programs. Give your observation and analysis for each of the problems.

1.

This program will display the text “OFFICIAL RECEIPT” but in different location, text color and background color. To change the text color to yellow the program set the text intensity to 6 so it turned yellow, the program also combined 3 different intensity colors to change the background to white, which are red, blue and green.

2.



This program used the function "funct" multiple times in a looping statement.

3.

```
1 3 6 10 15
```

This program uses addition rather than multiplication, despite including a loop and a function.

4.

```
My true north
Tara my love
The length og string name is 13
The length og string crush is 12
The new content of string name: My true northTara my love
The new length of string name is 25
Tara my love of my life.
Press any key to continue . . .
```

After determining the total number of characters in the inputs "My true north" and "Tara my love" (including the spaces), the program will output the phrase "Tara my love of my life."

5.

```
Enter any word/paragraph: I love Programming kahit bagsak
The reverse form is: kasgab tihak gnimmargorP evol I
Upper case letters: 2
Lower case letters : 25
Number : 0
Specialcharacters : 4
All caps: I LOVE PROGRAMMING KAHIT BAGSAK
Press any key to continue . . .
```

The reverse, number of digits, and special characters of the encoded word are all shown by the program. as well as the combined total of capital and lowercase characterslove of my life."

6.

```
20
```

The program is replicated ten times on a 1 by 10 array. Having the outcome show up as 20.

7.

```
19 20 21
23 24 25
```

A 3x2 array is used to generate values in this program, which are then shown with looping method.my life."

IV. Supplementary Problems

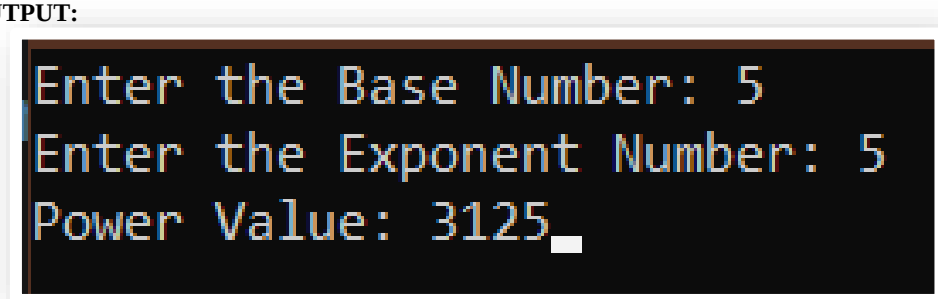
1. Write a function-oriented program that calculates the power value of the input base number and exponent number. Then display the power value. Write your input and output display statements at the main function and the process statement at the other function.

   Sample output:

   Enter the base number: 5
   Enter exponent number: 3
   Power value: 125

```
#include<iostream>
#include <conio.h>
using namespace std;
int power(int bNum, int eNum) {
	int res = 1;
	for (int i = 1; i <= eNum; i++)
		res = res * bNum;
	return res;
}
int main()
{
	int bNum, eNum, res;
	cout << "Enter the Base Number: ";
	cin >> bNum;
	cout << "Enter the Exponent Number: ";
	cin >> eNum;
	res = power(bNum, eNum);
	cout << "Power Value: " << res;
	_getch();
	return 0;
		}
```

OUTPUT:



2. Write a program using 3x4 array that searches a number and displays the number of times it occurs on the list of 12 input values.

Sample output:

Enter 12 numbers: 15 20 13 30 35 40 16 18 20 18 20 20
Enter a number to search: 20
Occurrence(s): 4

```
#include <iostream>
#include <conio.h>
using namespace std;

int main()
{
   int num[3][4];
   int numEnt;
   int occu{};
   cout << "Enter 12 numbers" << endl;
 for (int x = 0; x < 3; x++) {
      for (int i = 0; i < 4; i++) {
        cout <<  "[" << x << "][" << i << "]: ";
          cin >> num[x][i];}
   } for (int x = 0; x < 3; x++) {
     for (int i = 0; i < 4; i++) {
     cout << num[x][i] << "\t";}
   } cout << "\nEnter a number to search: ";
   cin >> numEnt;
   for (int x = 0; x < 3; x++) {
   for (int i = 0; i < 4; i++) {
   if (num[x][i] == numEnt) {
      occu++;}
      }
   } cout << "Occurence(s): " << occu;
   _getch();
   return 0;
}
```

OUTPUT:

3. Write a program using string functions that accepts a price of an item and display its coded value. The base of the key is:

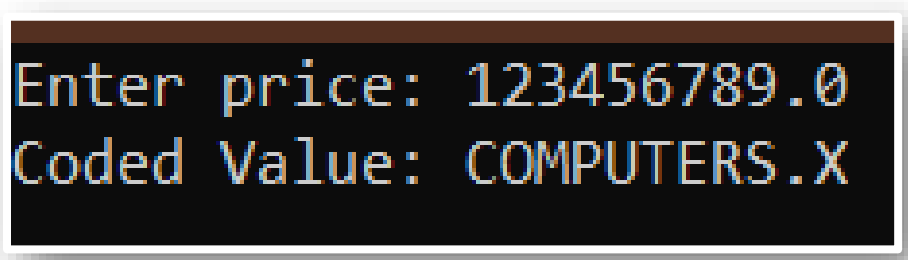
X	C	O	M	P	U	T	E	R	S
0	1	2	3	4	5	6	7	8	9

Sample output:
Enter price: 489.50
Coded value: PRS.UX

```
#include <iostream>
#include <conio.h>
using namespace std;

void Code(string input) {
   char pCodes[] = "XCOMPUTERS";
cout << "Coded Value: ";
   for (int i = 0; i < input.size(); i++)
   {if (input[i] == '.') {
cout << input[i];
} else {
cout << pCodes[input[i] - 48];
      }
   }
}
int main() {
   string input;
   cout << "Enter price: ";
   cin >> input;
   Code(input);
   _getch();
   return 0;
            }
```

OUTPUT:



V. Conclusion

I learned that arrays permit the declaration of several data element types. Arrays of all numeric types can be worked on and implemented equally, however arrays of all string types are distinctive. A function is an operation that may be encoded into a block of code. Input parameters are defined for a function so that callers can provide arguments.